

Portfolio Insights Generator - Take-Home Assignment

Time Limit: 72 hours **Position:** Intern - Software Engineering Intern

Overview

Build a Python application that processes financial transaction data, analyzes trading patterns, and uses AI to generate insights. This is a freeform assignment - you'll design the architecture and API yourself.

You're welcome to use AI tools (ChatGPT, Claude, Copilot, etc.) to help you. Just document what you used in your [SOLUTION.md](#) file.

What You'll Build

Three components:

1. **Transaction Processor** - Parse and analyze trading data
2. **Insights Generator** - Use LLM APIs to generate natural language insights
3. **Simple Dashboard** - Display data and insights with tool of your choice (e.g Streamlit)

The Data

Download here: Transactions Data

You'll work with **sample_transactions.csv** containing

- timestamp - When the transaction occurred
- ticker - Stock symbol (AAPL, GOOGL, etc.)
- action - BUY or SELL
- quantity - Number of shares
- price - Price per share
- trader_id - Trader identifier (T001, T002, etc.)

This is real-world financial data, so treat it as you would any data you receive in a production environment

Requirements

Part 1: Transaction Processing

Build a system to process and analyze transaction data:

Data Loading:

- Read the CSV file
- The data may have quality issues - identify and handle them appropriately
- Document any data cleaning decisions you made in [SOLUTION.md](#)

Data Storage and Retrieval:

- Store transactions so you can:
 - Quickly look up transactions by ticker symbol
 - Retrieve transactions in order (by time)
 - Handle 1000+ transactions efficiently

- You design the API - what functions/methods make sense?

Analytics: Calculate statistics on the data (include the following but feel free to add more):

- Total volume traded per ticker
- Net position per ticker (total buys - total sells)
- Most active traders (by transaction count)
- Basic time-based analysis

Part 2: LLM Integration

Integrate an LLM API to generate insights from the data:

API Integration:

- Choose OpenAI, Anthropic, Llama or even local llms (whichever you prefer)

Generate Insights:

- Use the LLM to identify patterns, risks, and unusual activity
- Example prompt ideas:
 - “Analyze these trading patterns...”
 - “Identify any concentration risks...”
 - “Flag unusual trading activity...”

Part 3: Dashboard

Create a simple dashboard with the tool of your choice (e.g. Streamlit) (UI/UX is NOT the focus):

Basic Requirements:

- Load and display the CSV data
- Show key statistics that you calculated:
 - Total transactions
 - Top tickers by volume
 - Most active traders
- Button to “Generate Insights” using your LLM integration
- Display the generated insights

Keep it simple - we’re evaluating your backend logic, not your design skills!

What to Submit

1. Your Code:

- All Python files (you choose the structure and naming)
- requirements.txt with all dependencies
- [README.md](#) with setup instructions specific to your implementation

2. [SOLUTION.md](#) (Important): This is where you explain your thinking and assumptions.

Use the provided SOLUTION_TEMPLATE.md as a guide. Include:

- **Data Processing Decisions:** What issues did you find? How did you handle them?
- **API Design:** How did you organize your code?
- **AI Tools Used:** What tools did you use and for what?
- **Time Spent:** Approximately how long this took

Evaluation Criteria

We'll evaluate based on:

- Core Python Fundamentals (45%)
- Problem Solving (25%)
- Integration & Functionality (20%)
- Communication (10%)

Follow-Up Interview

After you submit, we'll have a 45-minute technical interview where we'll discuss:

- Your design decisions
- Your data structure choices
- How your code works
- How you'd extend or modify the solution
- Your experience using AI tools (if applicable)

Good luck and we are looking forward to seeing how you approach this problem.